



Web Attacks & Data Privacy

Course Lectures (reminder)

1. Short intro	13.3 (extra)
2. Course overview: focus on terminology and ecosystem	27.3
3. Web Attacks Architecture	10.4
4. SQL Injection Attacks	17.4
5. Denial of Service (DoS)	24.4
6. Mitigating Web and Network Attacks - Security Tools and Techniques	1.5
7. Project definition – choosing theme & overview (hand-in before 23.5 EOD)	8.5 (15.5)
8. Cross Site Scripting (XSS) and Server Side Request Forgery (SSRF) comments	29.5
9. Mitigating Web and Network Attacks – continued comments	5.6
10. Man-in-the-Middle (MitM) presentation	12.6
11. Cross Site Request Forgery (CSRF), Social Engineering presentation	19.6
12. Testing for Security, Fingerprinting presentation	26.6
13. Data Privacy	28.6 (+)
14. Additional attacks – 2 nd layer attacks, NoSQL Injection, Buffer ovf, etc.	optional

4	5	6	7	8	9	10	11	12	13
S Q L	D O S	M T G	Theme Proj Demo	XSS SSRF	M T G	MiTM	CSRF Social	Testing, Fprinting	Data privacy
Ernan Bineth © Formulate project 23.5						Present	Present	Present	Handin 12.7.2026



Web Attacks & Data Privacy

Projects would c

- Simulate a vu
 - to SQLi
 - to web
 - to OWAS
- Monitoring a
- General Firev
- Monitoring a
- A LAN analys
- HoneyPot, di
- A website de

1. SQL injection – semi / non blind w/ automation
2. Denial of Service (DOS)
3. Man-in-the-Middle (MitM) and 2nd layer attacks
4. Web attacks website simulation (combination)
 - Cross Site Scripting (XSS)
 - Server Side Request Forgery (SSRF)
 - Cross Site Request Forgery (CSRF)
 - Social Engineering
 - OWASP main vulnerabilities
 - Buffer overflow
5. Mitigating Web and Network Attacks
 - Firewall, WAF
 - VPN
 - Honeypot
 - Intrusion Detection / Prevention Systems
 - LAN analyzer tool package
 - Anti-virus
 - Sandbox
6. Data Privacy
7. Others

ce, that would:
oopholes:

*Other ideas are
welcome...*

, data-link layer)

*No more than 2
proposals in
same domain
FCFS*



Web Attacks & Data Privacy

	theme	content
1	FW-DDOS	Monitoring & defending against inline Syn Flood and UDP Flood – identifying abnormalities using rate-limiting, reverse path filtering, heuristic stats and ML and blacklisting and log
2	WAF	Inline WAF demonstrating applicative defense against SQL Injection, XSS, Malicious File Uploads, DNS Amplification attacks and Slowloris using ML-Based Anomaly Detection
3	SQLi	2 URL based non blind script-based attacks – dictionary and union based, and 1 auto blind attack attempting log-in and data grab
4	LAN analyzer	LAN analyzer and basic vulnerability assessment of transmission in LAN devices. Port scanner NMAP, fingerprinting p0f, Scapy traffic analyzer ARP spoofing, whitelisted IP, filtering, alert
5	MitM	Addressing (1) ARP spoofing to change an incoming DNS response to poison DNS client cache (2) DHCP spoofing – attack & output, detection and prevention.
6	Honey pot	Router, asset and HP detecting and routing SQLi and Brute-force on credential-based data web-page. whitelist/blacklist
7	noSQLi	Demonstrating noSQL attack using burp-suite vs python (1) brute forcing password / access product data (2) comparing python and semi blind automation finding info in tables
8	XSS	XSS Persistent stealing cookie by ascii input field, URL SSRF access forbidden files, script crawl identify & log vulnerabilities & remedy



Web Attacks & Data Privacy

Course Assignment proposal – presentation slides **hand-in 23/5**

Hand in moodle

1. Heading – assignment topic and team members (names, ID)
2. Rational – the main objectives & functionality the implementation must fulfill
3. Course link – what topics of the course are mapped in the implementation
4. Leading products in the market – description, advantages / disadvantages
5. Topological diagram – building blocks: components, cloud, databases, modules
6. Technology – OS, languages, tools, DB, libraries, rational for choice.
7. Assumptions and constraints – what is assumed for your project to work
8. main scenarios and expected result
9. Challenges – main implementational and technical challenges & mitigation
10. Main algorithms – training ML, statistics, user input validation,...
11. High level planning – separation of roles and milestones



Web Attacks & Data Privacy

Final handout **hand-in 12/7/26 gmail**

Submit by one of the team members by Sunday night **12.7** EOD the following:

- **Wireshark pcap** file including all scenario packets
- **Clip** up to 6 min, **narrated equally** by both team members, demonstrating:
 - The **topology & naïve scenario** – applications per hosts, Wireshark recording spot
 - Each **scenario** – attack, capture, analysis, decision, result, mitigate (if relevant)
 - Main algorithms, rule-base data, simulation parameters (Define used libraries) on attack, analysis and defense
- **Documented code**
- **Updated presentation** matching the clip, including:
 - Updated topology with participating IPs in the WS recording per role (match WS)
 - Detailed walkthrough on each **scenario** reflecting **Wireshark records**

Project in pairs, max two projects in the same domain – FCFS (inform ASAP)

Your language of choice (English/Hebrew, C# / Java / JS / Python). Absolutely original.

Requires approval in class presentation of proposal (**20%**) before implementation

Sent by due-date - code and sample run (**70%**) & (personal) demonstrated in video (**10%**)